

As established in Chapter 2, the foundation of portfolio selection lies in Harry Markowitz's Modern Portfolio Theory (MPT), which formalizes the risk-return trade-off by assuming normally distributed returns and relying on symmetric variance as the primary measure of risk (Markowitz, 1952).

However, the empirical reality of emerging markets like the Nigerian Exchange Group (NGX) fundamentally violates these strict assumptions; these markets are instead characterized by high volatility, low liquidity, asymmetric information, and leptokurtic distributions where extreme downside market events happen much more frequently than normal distributions would predict (Arif & Sohail, 2020). To address these structural violations, Post-Modern Portfolio Theory (PMPT) offers a theoretical solution by redefining risk not as general symmetrical volatility, but specifically as the probability of financial loss, thereby distinguishing between desirable upside gains and detrimental downside drops (Alotaibi et al., 2022). Consequently, this methodology operationalizes PMPT by discarding the traditional mean-variance framework in favor of Conditional Value-at-Risk (CVaR), a coherent risk measure selected to specifically quantify and minimize the average severity of extreme tail losses occurring beyond a catastrophic threshold in the NGX (Rockafellar & Uryasev, 2002).

3.1.2 Machine Learning Theories for Forecasting

To address the limitations of the Efficient Market Hypothesis (EMH) outlined in Chapter 2, this methodology utilizes advanced statistical learning theories. The EMH and the associated Random Walk Hypothesis postulate that asset prices instantaneously reflect all available information and fluctuate randomly, theoretically rendering future price movements unpredictable and active forecasting futile (Fama, 1970).

However, substantial empirical evidence challenges these strict assumptions, particularly in emerging markets like the Nigerian Exchange Group (NGX), which exhibit market frictions,

significant volatility clustering, frequent structural breaks, and complex non-linear correlations (Adegboyo & Sarwar, 2025). Statistical learning theories offer a robust mathematical framework that bypasses rigid econometric assumptions, allowing for the extraction of meaningful, latent non-linear dependencies directly from noisy and high-dimensional financial data (Gu, Kelly, & Xiu, 2019).

Consequently, this methodology adopts advanced ensemble machine learning algorithms—specifically Random Forest and eXtreme Gradient Boosting (XGBoost)—to effectively operationalize statistical learning, thereby capturing the complex, non-stationary market patterns that traditional models and the EMH fail to identify (Manogna & Kulkarni, 2025; Ashrafzadeh et al., 2025).

3.2 Model Specification

3.2.1 Predictive Models

The aim of the predictive phase of the study is to forecast future returns of the component assets within the NGX Pension Index. To achieve this, the study frames return forecasting as a supervised multivariate time-series problem. Unlike traditional linear asset pricing models which assume a linear relationship between risk factors and returns, this methodology posits that financial time series are driven by complex, non-linear, and interacting market dynamics.